

Project Update 5

ECON 345

Due: 11:59PM October 24, 2026

For this assignment, you will need to submit:

1. Your RMarkdown file (**download and use the template we provided, `project_hw5_student.Rmd`**)
2. A knitted PDF.

If you cannot knit to a PDF, type your answers into a Word document (remember to number them accordingly), and be sure to include screenshots of your outputs for each question. Submit this Word document together with your `.Rmd` file so we can later help troubleshoot your problem with knitting.

Part A - Iterating on Previous Work (30 points)

For this assignment, you need to reproduce your work from Project Update 4 with revisions based on feedback you received. If you did not receive any feedback for a particular item, you should still include that item.

1. Research Question (10 points)

- What is your research question? (2)
- How has your research question changed over the course of the project? (2)
- What are the key relationships in your data that could inform the answer to your research question? (2)
- Based on your data, what is a testable hypothesis for your research question? For example, if your research question was “How does education affect earnings?” and your dataset included variables for individuals’ years of education and annual salaries, you could propose the testable hypothesis that “Individuals with more years of education have, on average, higher annual salaries.” (4)

2. Summary Statistics (4 points)

- Using the `vtable` or `kable` package and incorporating weights (if your project has weights), compute and present summary statistics of your dependent and independent variables. (2)
- If you received feedback on your summary statistics or their interpretation, state the feedback and then incorporate it here. If there’s anything you’re struggling to incorporate, state that. If you had no feedback for this item, state that. (2)

3. Conditional Summary Statistics (4 points)

- Using the `vtable` or `kable` package and incorporating weights (if your project has weights), compute and present conditional summary statistics of your dependent variable by levels/groups of your independent variable(s). (2)
- If you received feedback on your conditional summary statistics or their interpretation, state the feedback and then incorporate it here. If there’s anything you’re struggling to incorporate, state that. If you had no feedback for this item, state that. (2)

4. Data Visualization (12 points)

- Create three charts that are key to telling your story. At least one of these should show a key relationship between two separate variables (e.g., a conditional histogram or scatterplot). Refer to the data visualization checklist provided with this assignment for tips on making your charts easy to read and informative. (5)
- For each chart, describe the part of the story it tells. (For example, your chart may show wages by gender, and it tells the story that wages vary substantially by gender.) (5)

- If you received feedback on your data visualizations or their interpretation, state the feedback and then incorporate it here. If there's anything you're struggling to incorporate, state that. If you had no feedback for this item, state that. (2)

Part B - Refining Your Interpretation (30 points)

5. What did you originally expect to find in the data about the relationship between your key variables? (5)
6. How have your findings differed from your original expectations? (5)
7. How does your analysis thus far help inform your research question? Tell us a story. (10)
8. What limitations exist to your analysis and visualizations so far? How could you overcome these limitations? (10)

Part C - Analysis With a T-Test (20 points)

9. Define a grouping variable based on your independent variable that splits your data into two groups. Explain your reasoning for how you've grouped the data. You may want to use the same groups as in Question 4. (5)
10. Based on your summary statistics or histogram, describe the relationship between your dependent and independent variables — positive, negative, or neither? (Hint: compare means.) (5)
11. Set up a test to examine the statistical significance of the result you found above.
 - Are you using a one-sample or two-sample t-test? One-sided or two-sided? (2)
 - State your null and alternative hypotheses. (3)
12. Run the t-test described above with `infer::t_test()` and interpret the results. (5)

Part D - Follow-up With Your Adviser (10 points)

13. Email your adviser with your knitted output (the PDF or Word document that results from your RMarkdown file). Do not send your R script or your .Rmd, but do submit those on Canvas.
 - Email your knitted output to your adviser, with `econ345.spelman@gmail.com` cc'd. (5)
 - In your email, give your adviser a brief summary of what you've done this week and what you've learned about your research question. Include any questions or ideas about where to take your analysis next. (5)

Part E - Share With Your Classmates (10 points)

14. Pick one chart you particularly like from Part A and share it with your classmates on the Canvas discussion board, with a brief description of what it means.
 - Posting your chart (5)
 - A brief description of your chart (5)

You're strongly encouraged to comment on your classmates' posts as well!